



**2008**

HIGHER SCHOOL CERTIFICATE  
TRIAL EXAMINATION

# Mathematics Extension 2

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start a new booklet for each question

## Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value

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**Total Marks – 120**

**Attempt Questions 1-8**

**All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

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**Question 1** (15 marks) Use a SEPARATE writing booklet.

**Marks**

(a)  $\int \frac{2x}{\sqrt{1-x^4}} dx$  2

(b)  $\int_0^{\frac{\pi}{2}} \frac{dx}{2+\cos x}$  3

(c)  $\int_1^{e^2} 3x^2 \ln x \, dx$  3

(d)  $\int \frac{dx}{\sqrt{x^2 - x + 1}}$  2

(e) (i) Show that  $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$  2

(ii) Use this property to show that  $\int_0^1 x^3(1-x)^6 \, dx = \frac{1}{840}$  3

**Question 2** (15 marks) Use a SEPARATE writing booklet.

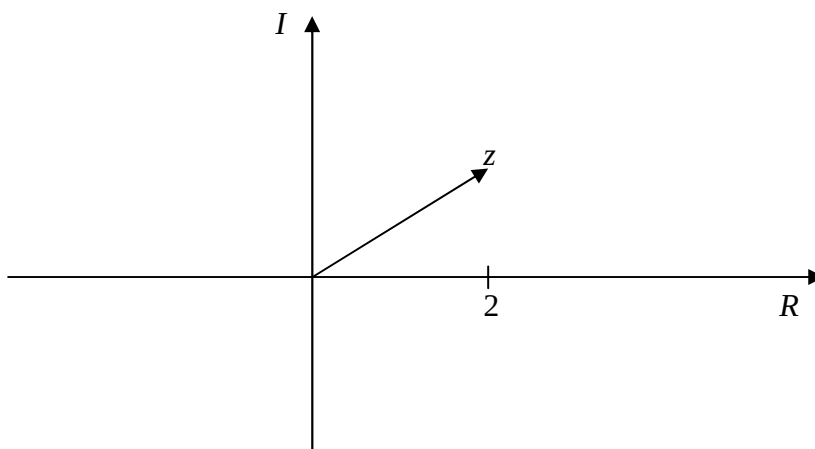
**Marks**

(a) A complex number  $z$  is given by  $z = \sqrt{3} + i$

(i) Evaluate  $\bar{z}$ . Verify that  $z\bar{z}$  is real. 2

(ii) Find  $\frac{1}{z}$  in the form  $a + ib$ , where  $a$  and  $b$  are real. 1

(b) A point  $z$  on the Argand Diagram is given below:



Copy this diagram into your examination booklet and use it to plot the following points. Clearly label each point.

(i)  $\bar{z}$  1

(ii)  $2iz$  1

(iii)  $\frac{1}{z}$  1

(c) Express  $i - 1$  in modulus argument form, and hence simplify  $(i - 1)^5$  2

**Question 2 continues on page 4**

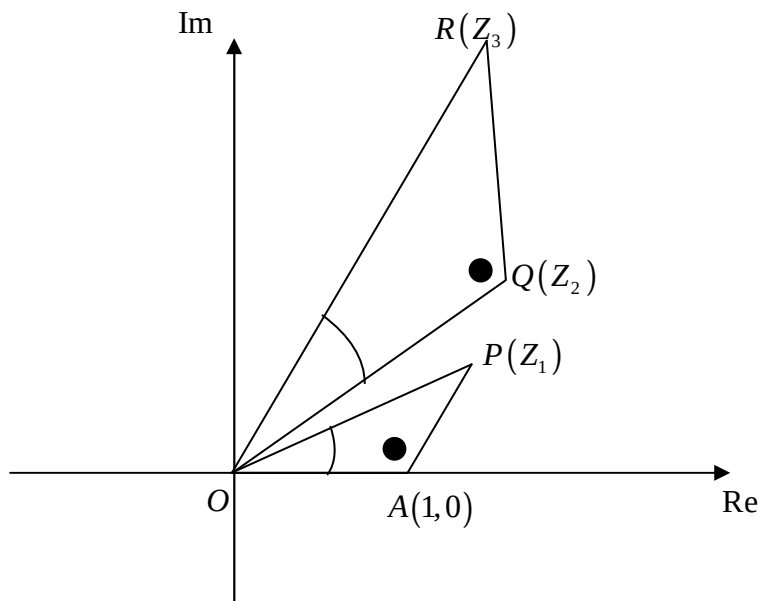
Question 2 (continued)

(d) Sketch the locus and state its equation:

(i)  $|z - 2| = |z - 2i|$  2

(ii)  $z\bar{z} - 3(z + \bar{z}) \leq 0$  2

(e)



In the figure above, the points  $P$ ,  $Q$  and  $A$  represent the complex numbers  $Z_1, Z_2$  and  $(1,0)$  respectively. Given  $\angle OAP = \angle OQR$  and  $\angle AOP = \angle QOR$ .

Explain why  $R(Z_3)$  represents the complex number  $Z_1 Z_2$ . 3

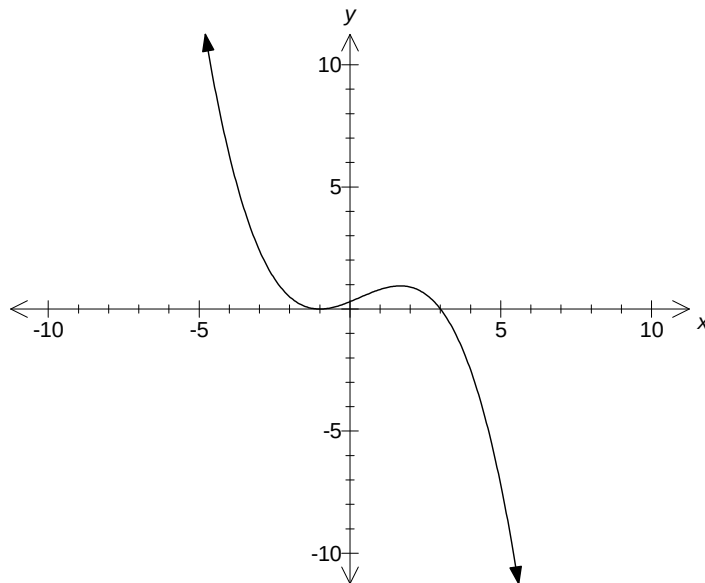
You must support your answer with clear and complete mathematical reasons.

**End of Question 2**

**Question 3** (15 marks) Use a SEPARATE writing booklet.

**Marks**

(a)



The graph of  $f(x) = \frac{1}{10}(x+1)^2(3-x)$  is drawn above.

On separate diagrams, draw a neat sketch showing the main features of each of the following:

- |       |                  |          |
|-------|------------------|----------|
| (i)   | $y = f(x-1)$     | <b>1</b> |
| (ii)  | $y = f( x )$     | <b>1</b> |
| (iii) | $y = \{f(x)\}^2$ | <b>2</b> |
| (iv)  | $y = xf(x)$      | <b>2</b> |
| (v)   | $y^2 = f(x)$     | <b>2</b> |
| (vi)  | $y = e^{f(x)}$   | <b>2</b> |

(b) Given that  $I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$  show that :

- |      |                                                                      |          |
|------|----------------------------------------------------------------------|----------|
| (i)  | $I_n = \frac{1}{n-1} \left( (\sqrt{2})^{n-2} + (n-2)I_{n-2} \right)$ | <b>4</b> |
| (ii) | Hence or otherwise evaluate $I_4$                                    | <b>1</b> |

**End of Question 3**

**Question 4** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- (a) (i) If  $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ , show that  $P(x) = 0$  has a multiple root, find this root and its multiplicity. **3**
- (ii) Hence factorise  $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$  into its linear factors. **1**
- (b) The equation  $x^3 + 2x - 1 = 0$  has roots  $\alpha, \beta, \gamma$ . Find the monic equations with roots
- (i)  $\alpha^2, \beta^2, \gamma^2$ . **2**
- (ii)  $\alpha\beta, \beta\gamma, \alpha\gamma$  **3**
- (iii) Evaluate  $\alpha^3 + \beta^3 + \gamma^3$  **2**
- (c) A point  $P\left(ct, \frac{c}{t}\right)$  lies on the rectangular hyperbola  $xy = c^2$ .
- (i) Show that the equation of the tangent at the point  $P\left(ct, \frac{c}{t}\right)$  on the rectangular hyperbola is given by  $x + t^2y = 2ct$ . **2**
- (ii) Prove that the area bounded by the tangent and the asymptotes of the rectangular hyperbola is a constant. **2**

**End of Question 4**

**Question 5** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- (a)  $ABC$  is an equilateral triangle, inscribed in a circle.  $X$  is a point on the minor arc  $BC$ .

(i) Prove that  $\triangle BD X \parallel \triangle AC X$

**3**

(iii) Prove that  $XB + XC = XA$

**3**

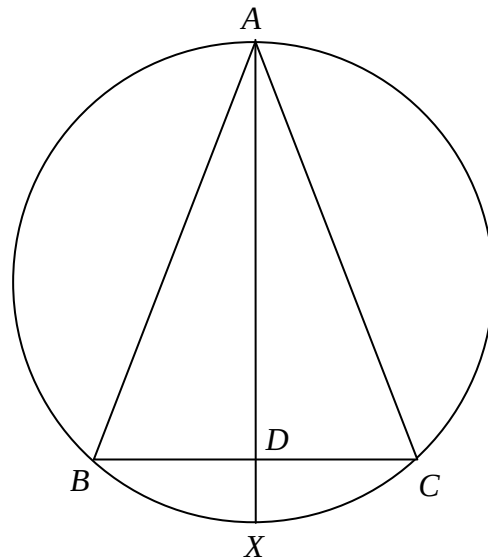


DIAGRAM NOT TO SCALE

- (b) State whether each of the following are true or false giving brief reasons for your answers:

(i)  $\int_0^{\pi} \sin 9x \, dx = 0$

**1**

(ii)  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin x \, dx = 0$

**2**

- (c) Find the equation of the tangent to the curve  $\cos 2x + \sin y = 1$  at the point  $x = \frac{\pi}{6}$ .

**3**

- (d) Use the substitution  $x = a \sin \theta$  to show that

**3**

$$\int \sqrt{a^2 - x^2} \, dx = \frac{1}{2} a^2 \sin^{-1} \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} + C$$

**End of Question 5**

**Question 6** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- (a) (i) Given  $a\alpha^2 + b\alpha + c = 0$  where  $a, b, c \in \mathbb{R}$  and  $\alpha \in \mathbb{C}$ , prove that  $a(\overline{\alpha})^2 + b\overline{\alpha} + c = 0$  2
- (ii) A polynomial  $P(x)$  with real coefficients, has two of its zeros  $3i$  and  $1 + 2i$ . Find in expanded form, a possible polynomial  $P(x)$ . 3
- (b) Use De Moivre's Theorem and binomial expansion to find an expression for  $\cos 4\theta$  in terms of  $\cos \theta$ . 3
- (c) (i) Given  $z = \cos \theta + i \sin \theta$ , prove  $z^n + \frac{1}{z^n} = 2 \cos n\theta$  2
- (i) Hence by considering the expansion  $\left(z + \frac{1}{z}\right)^4$  show that 3
- $$\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$$
- (iii) Hence evaluate  $\int_0^{\frac{\pi}{2}} \cos^4 \theta \, d\theta$  2

**End of Question 6**

**Question 7** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- (a) The roots of the polynomial  $p(x) = x^3 + ax^2 + bx + c = 0$  are three consecutive terms of an arithmetic series. Prove that the relationship between the coefficients is given by  $2a^3 - 9ab + 27c = 0$

**4**

Hint: make an appropriate choice for the roots in arithmetic progression.

- (b) A point  $P(a \cos \theta, b \sin \theta)$  lies on the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  where  $a > 0$  and  $b > 0$ .

The equation of the normal at the point  $P(a \cos \theta, b \sin \theta)$  is given by

$$xa \sin \theta - yb \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

- (i) Show that the ellipse intersects the rectangular hyperbola  $xy = c^2$  in four points if  $ab > 2c^2$

**3**

- (ii) Show that for  $0 < \theta < \frac{\pi}{2}$ , the normal at  $P$  on the ellipse intersects the hyperbola in two distinct points, say  $A$  and  $B$ .

**3**

- (iii) If  $M$  is the mid-point of  $AB$ , show that the coordinates of  $M$  are given by

**2**

$$\left( \frac{(a^2 - b^2) \cos \theta}{2a}, -\frac{(a^2 - b^2) \sin \theta}{2b} \right)$$

- (iv) Hence find the locus of  $M$  as  $\theta$  varies.

**3**

**End of Question 7**

**Question 8** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- (a) For the function  $y = \cos^{-1}(e^x)$ ,
- (i) Find the domain and the range. 2
  - (ii) Draw a neat sketch the graph of  $y = \cos^{-1}(e^x)$ . 2
  - (iii) Hence draw a neat sketch of the curve  $y = \frac{1}{(\cos^{-1}(e^x))}$  2
- (b) (i) Using induction, show that for each positive integer  $n$ , there are unique positive integers  $p_n$  and  $q_n$  such that:  $(1+\sqrt{2})^n = p_n + q_n\sqrt{2}$  4
- (ii) Show also that  $p_n^2 - 2q_n^2 = (-1)^n$ . 1
- (c) If  $f(xy) = f(x) + f(y)$ , for all  $x, y \neq 0$ , prove that
- (i)  $f(1) = f(-1) = 0$  2
  - (ii)  $f(x)$  is an even function. 2

**End of paper**

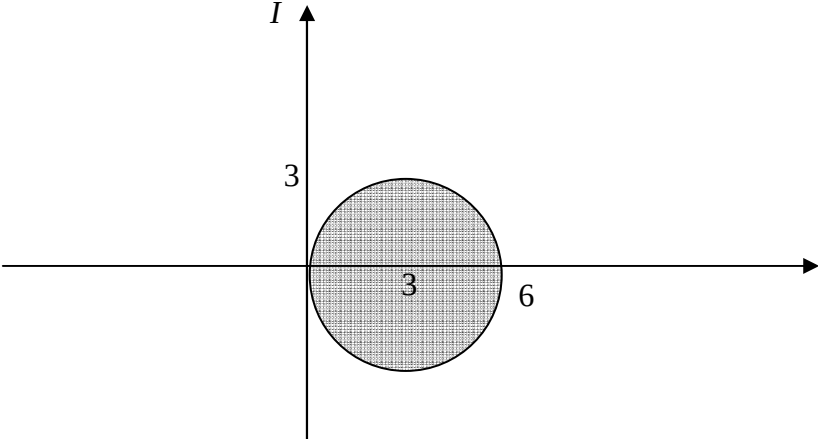
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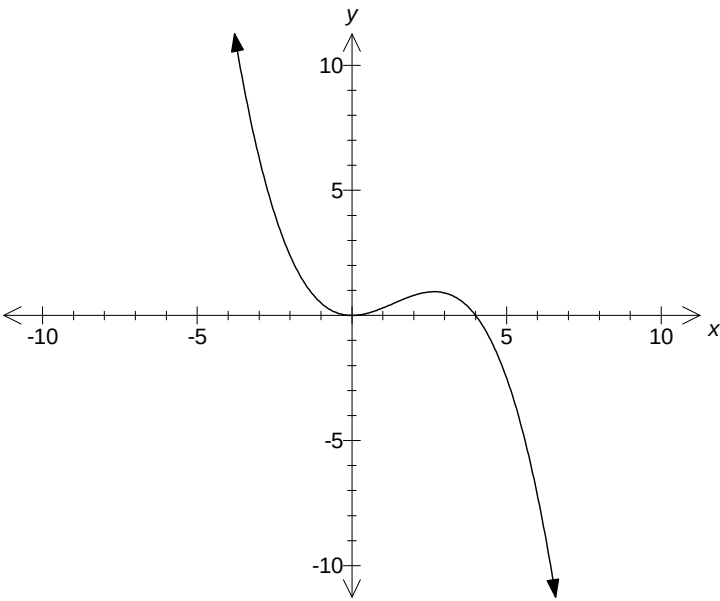
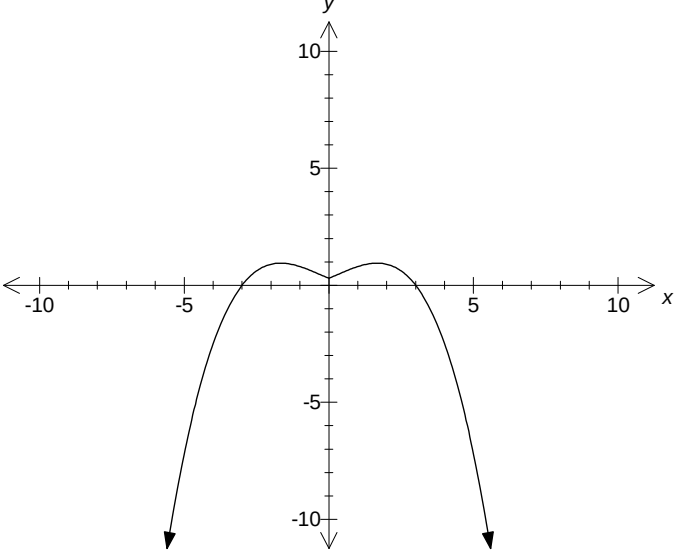
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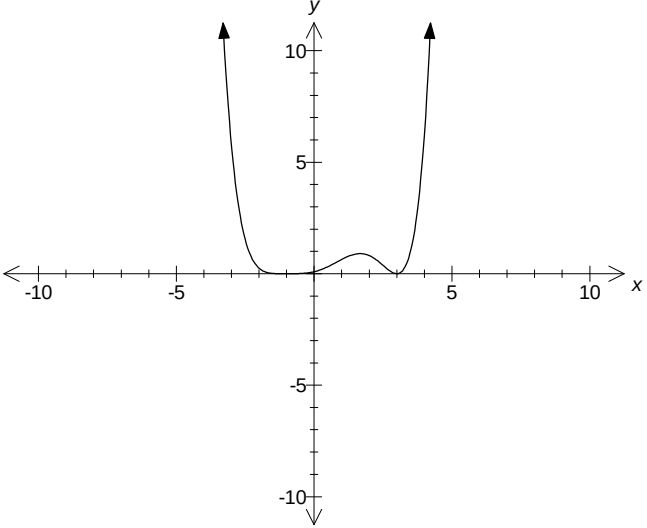
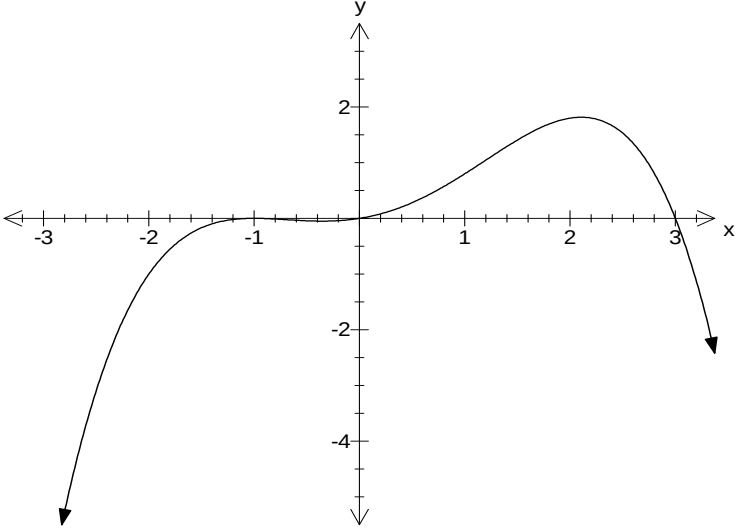
| Question | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks | Bands |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| 1(a)     | $\int \frac{2x}{\sqrt{1-x^4}} dx \quad \text{Let } u = x^2 \quad \therefore \frac{du}{dx} = 2x \quad \text{or} \quad dx = \frac{du}{2x} \quad \boxed{\checkmark}$ $\therefore \int \frac{2x}{\sqrt{1-x^4}} dx = \int \frac{2x}{\sqrt{1-u^2}} \cdot \frac{du}{2x}$ $= \int \frac{1}{\sqrt{1-u^2}} du$ $= \sin^{-1} u + C$ $= \sin^{-1} x^2 + C \quad \boxed{\checkmark}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |       |
| 1(b)     | $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$ $\text{Let } t = \tan \frac{x}{2}$ $\therefore \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} = \frac{1}{2} [1 + \tan^2 \frac{x}{2}] = \frac{1}{2} [1 + t^2] \quad \text{or} \quad dx = \frac{2 dt}{1+t^2}$ $\text{and } \cos \theta = \frac{1-t^2}{1+t^2} \quad \boxed{\checkmark}$ $\therefore \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x} = \int_0^1 \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2 dt}{1+t^2}$ $\text{since } t = \tan \frac{\frac{\pi}{2}}{2} = 1 \quad \text{and} \quad t = \tan \frac{0}{2} = 0$ $= \int_0^1 \frac{1}{\frac{2(1+t^2)}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2 dt}{1+t^2}$ $= \int_0^1 \frac{1+t^2}{3+t^2} \cdot \frac{2 dt}{1+t^2} \quad \boxed{\checkmark}$ $= \int_0^1 \frac{2}{3+t^2} dt$ $= 2 \int_0^1 \frac{1}{3+t^2} dt$ $= 2 \left[ \frac{1}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 \quad \boxed{\checkmark}$ $= \frac{\pi}{3\sqrt{3}}$ | 3     |       |

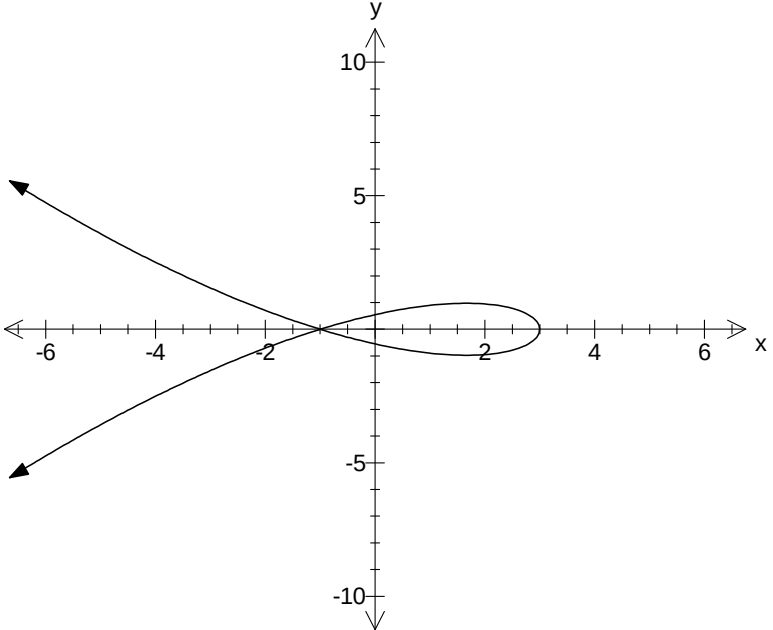
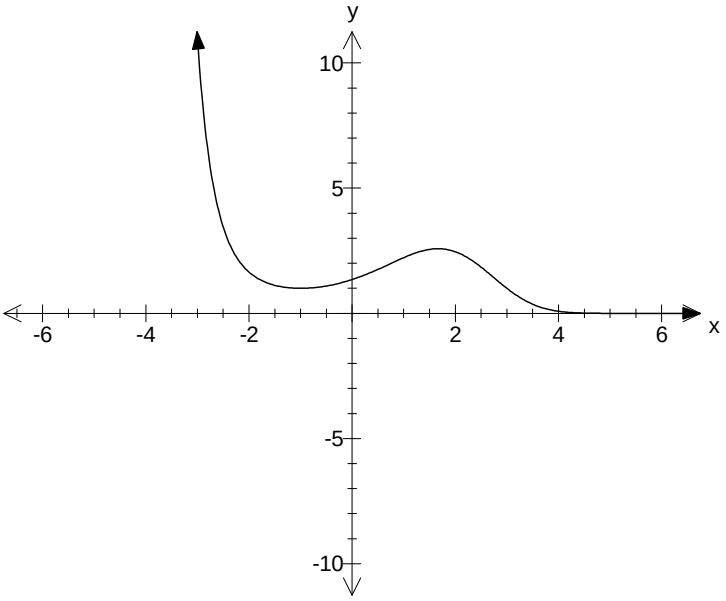
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|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 1(c)     | $\int_1^{e^2} 3x^2 \ln x \, dx \quad \text{Let } u = \ln x \quad \frac{dv}{dx} = 3x^2 \quad \frac{du}{dx} = \frac{1}{x} \quad v = x^3$ $\therefore \int_1^{e^2} 3x^2 \ln x \, dx = uv - \int v \, du \quad \checkmark$ $= \left[ x^3 \ln x \right]_1^{e^2} - \int x^3 \frac{dx}{x}$ $= \left[ x^3 \ln x \right]_1^{e^2} - \int x^2 \, dx$ $= \left[ x^3 \ln x \right]_1^{e^2} - \left[ \frac{x^3}{3} \right]_1^{e^2} \quad \checkmark$ $= \left[ e^6 \ln e^2 - 1^3 \ln 1 \right] - \left[ \frac{e^6}{3} - \frac{1}{3} \right]$ $= 2e^6 - \frac{e^6}{3} + \frac{1}{3}$ $= \frac{5e^6 + 1}{3} \quad \checkmark$ | 3 |  |
| 1(d)     | $\int \frac{dx}{\sqrt{x^2 - x} + 1} = \int \frac{1}{\sqrt{x^2 - x + (\frac{1}{2})^2 + 1 - (\frac{1}{2})^2}} \, dx \quad \checkmark$ $= \int \frac{1}{\sqrt{(x - \frac{1}{2})^2 + \frac{3}{4}}} \, dx$ $= \log \left  (x - \frac{1}{2}) + \sqrt{(x - \frac{1}{2})^2 + \frac{3}{4}} \right  + C \quad \checkmark$                                                                                                                                                                                                                                                                                               | 2 |  |
| 1(e)(i)  | $\int_0^a f(a-x) \, dx \quad \text{Let } u = a-x \quad \therefore \frac{du}{dx} = -1$ $\text{If } x=a \text{ then } u=a-a= \quad x=0 \text{ then } u=a-0=a \quad \checkmark$ $\int_0^a f(u) \cdot -du = - \int_a^0 f(u) \, du$ $= \int_0^a f(u) \, du \quad \checkmark$                                                                                                                                                                                                                                                                                                                                       | 2 |  |
| 1(e)(ii) | $\int_0^1 x^3(1-x)^6 \, dx = \int_0^1 (1-x)^3(1-(1-x))^6 \, dx \quad \checkmark$ $= \int_0^1 (1-3x+3x^2-x^3)x^6 \, dx$ $= \int_0^1 x^6 - 3x^7 + 3x^8 - x^9 \, dx \quad \checkmark$ $= \left[ \frac{x^7}{7} - \frac{3x^8}{8} + \frac{x^9}{3} - \frac{x^{10}}{10} \right]_0^1 \quad \checkmark$ $= \frac{1}{840}$                                                                                                                                                                                                                                                                                               | 3 |  |



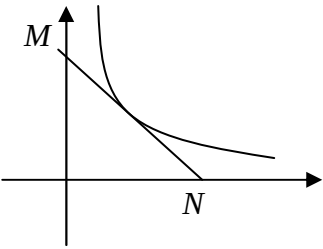
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|
| 2(d)(ii) |  <p><math>(x-3)^2 + y^2 \leq 9</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                   |  |
| 2(e)     | <p> <math>\angle AOR = \angle AOQ + \angle QOR</math><br/> <math>= \angle AOQ + \angle AOP</math><br/> <i>i.e.</i> <math>\arg z_3 = \arg z_2 + \arg z_1</math><br/> the triangles <math>ORQ</math> and <math>OPA</math> are equiangular and hence similar<br/> <math>\therefore</math> their sides are proportional<br/> <math display="block">\frac{OR}{OP} = \frac{OQ}{OA} = \frac{ z_3 }{ z_1 } = \frac{ z_2 }{ 1 }</math> <math display="block"> z_3  = \frac{ z_2  z_1 }{1}</math> <math display="block">=  z_2  z_1 </math> <math>\therefore z_3 = z_2 \cdot z_1</math> <i>i.e.</i> <math>R</math> represents the complex number <math>z_2 z_1</math> </p> | 1<br><br>1<br><br>1 |  |

| Question | Criteria                                                                                                                                                              | Marks    | Bands |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|
| 3(a)(i)  |  <p><input checked="" type="checkbox"/> <i>shift graph 1 place to the right</i></p> | <b>1</b> |       |
| 3(a)(ii) |  <p><input checked="" type="checkbox"/> <i>reflection in y axis</i></p>             | <b>1</b> |       |

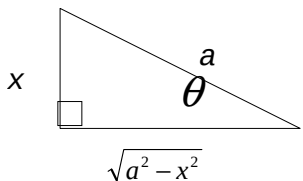
|           |                                                                                                                                                                                                                                                                                                                               |   |  |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 3(a)(iii) |  <p> <input checked="" type="checkbox"/> graph above x-axis<br/> <input checked="" type="checkbox"/> graph <math>0 &lt; y &lt; 1</math> (less steep gradient)<br/>             and graph <math>y &gt; 1</math> steeper gradient         </p> | 2 |  |
| 3(a)(iv)  |  <p> <input checked="" type="checkbox"/> 3 roots at -1, 0 and 3<br/> <input checked="" type="checkbox"/> 2 stationary points         </p>                                                                                                  | 2 |  |

|          |                                                                                                                                                                                                                                                                                                                                                                                          |   |  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 3(a)(v)  |  <p> <input checked="" type="checkbox"/> sketch only +ve section of original graph and reflect in <math>x</math> - axis<br/> <input checked="" type="checkbox"/> for <math>-1 &lt; y &lt; 1</math> the sketch is less steep and more steep after <math>y &lt; -1</math> and <math>y &gt; 1</math> </p> | 2 |  |
| 3(a)(vi) |  <p> <input checked="" type="checkbox"/> where cuts <math>x</math> - axis now becomes <math>(-1,1)</math> and <math>(3,1)</math><br/> <input checked="" type="checkbox"/> anything that was negative now becomes an asymptote above the <math>x</math> - axis.         </p>                          | 2 |  |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 3(b)(i)  | $I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$ $\int \sec^n x \, dx = \int \sec^{n-2} x \cdot \sec^2 x \, dx \quad \checkmark$ <p>where <math>u = \sec^{n-2} x \quad \frac{dv}{dx} = \sec^2 x</math></p> $\frac{du}{dx} = (n-2)\sec^{n-3} x \sec x \tan x \quad v = \tan x$ $\int \sec^{n-2} x \cdot \sec^2 x \, dx = uv - \int v \, du$ $= \sec^{n-2} x \tan x - \int (n-2)\sec^{n-3} x \sec x \tan^2 x \, dx$ $= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \tan^2 x \, dx$ $= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) \, dx$ $= \sec^{n-2} x \tan x - (n-2) \int \sec^n x - \sec^{n-2} x \, dx \quad \checkmark$ $\therefore \int \sec^n x \, dx = \sec^{n-2} x \tan x - (n-2) \int \sec^n x - (n-2) \int \sec^{n-2} x \, dx$ $\int \sec^n x \, dx + (n-2) \int \sec^n x = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \, dx$ $(n-1) \int \sec^n x \, dx = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \, dx$ $\int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x - \frac{n-2}{n-1} \int \sec^{n-2} x \, dx \quad \checkmark$ $I_n = \frac{1}{n-1} \left[ \left[ \sec^{n-2} x \tan x \right]_0^{\frac{\pi}{4}} - (n-2) I_{n-2} \right]$ $I_n = \frac{1}{n-1} \left[ \left[ \sec^{n-2} \left( \frac{\pi}{4} \right) \tan \left( \frac{\pi}{4} \right) - \sec^{n-2} (0) \tan(0) \right] - (n-2) I_{n-2} \right] \quad \checkmark$ $I_n = \frac{1}{n-1} \left[ \left( \sqrt{2} \right)^{n-2} - (n-2) I_{n-2} \right]$ | 4 |  |
| 3(b)(ii) | $I_4 = \frac{1}{3} \left[ \left( \sqrt{2} \right)^2 - 2I_2 \right]$ $I_2 = \frac{1}{2} \left[ \left( \sqrt{2} \right)^0 - 0I_0 \right] = \frac{1}{2}$ $\therefore I_4 = \frac{1}{3} \left[ \left( \sqrt{2} \right)^2 - 2 \left( \frac{1}{2} \right) \right]$ $= \frac{2}{3} - \frac{1}{3}$ $= \frac{1}{3} \quad \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 |  |

| Question  | Criteria                                                                                                                                                                                                                                                                                                                                  | Marks                                                                                                  | Bands |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------|
| 4(a)(i)   | $P'(x) = 4x^3 + 3x^2 - 6x - 5$<br>$P''(x) = 12x^2 + 6x - 6$<br>$12x^2 + 6x - 6 = 0 \Rightarrow x = -1, -\frac{1}{2}$<br>$P'(-1) = 0$<br>$\therefore x = -1$ is a root of multiplicity 3                                                                                                                                                   | <p>1</p> <p>1</p> <p>1</p>                                                                             |       |
| 4(a)(ii)  | $x^4 + x^3 - 3x^2 - 5x - 2 = (x+1)^3(x-2)$                                                                                                                                                                                                                                                                                                | 1                                                                                                      |       |
| 4(b)(i)   | $(\sqrt{x})^3 + 2\sqrt{x} - 1 = 0$<br>$x\sqrt{x} + 2\sqrt{x} = 1$<br>$(\sqrt{x}(x+2))^2 = 1$<br>$x(x^2 + 4x + 4) = 1$<br>$x^3 + 4x^2 + 4x - 1 = 0$                                                                                                                                                                                        | 1                                                                                                      |       |
| 4(b)(ii)  | $\alpha\beta, \alpha\gamma, \beta\gamma = \frac{\alpha\beta\gamma}{\gamma}, \frac{\alpha\beta\gamma}{\beta}, \frac{\alpha\beta\gamma}{\alpha} = \frac{1}{\gamma}, \frac{1}{\beta}, \frac{1}{\alpha}$<br>$\left(\frac{1}{x}\right)^3 + 2\left(\frac{1}{x}\right) - 1 = 0$<br>$\frac{1}{x^3} + \frac{2}{x} - 1 = 0$<br>$x^3 - 2x^2 - 1 = 0$ | 1<br>1<br>1                                                                                            |       |
| 4(b)(iii) | $\alpha^3 + 2\alpha - 1 = 0$<br>$\beta^3 + 2\beta - 1 = 0$<br>$\gamma^3 + 2\gamma - 1 = 0$<br>$(\alpha^3 + \beta^3 + \gamma^3) + 2(\alpha + \beta + \gamma) - 3 = 0$<br>$\therefore \alpha^3 + \beta^3 + \gamma^3 = 3 - 2(0)$<br>$= 3$                                                                                                    | 1<br>1                                                                                                 |       |
| 4(c)(i)   | $y' = -\frac{c}{x^2}$ at $\left(ct, \frac{c}{t}\right)$ $y' = -\frac{1}{t^2}$<br>$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$<br>$yt^2 - ct = -x + ct$<br>$yt^2 + x = 2ct$                                                                                                                                                                  | 1<br>1                                                                                                 |       |
| 4(c)(ii)  | $M : x = 0 \Rightarrow y = \frac{2c}{t}$<br>$N : y = 0 \Rightarrow x = 2ct$<br>$\text{area} = \frac{1}{2}(2ct)\left(\frac{2c}{t}\right)$<br>$= 2c^2$ which is a constant                                                                                                                                                                  |  <p>1</p> <p>1</p> |       |

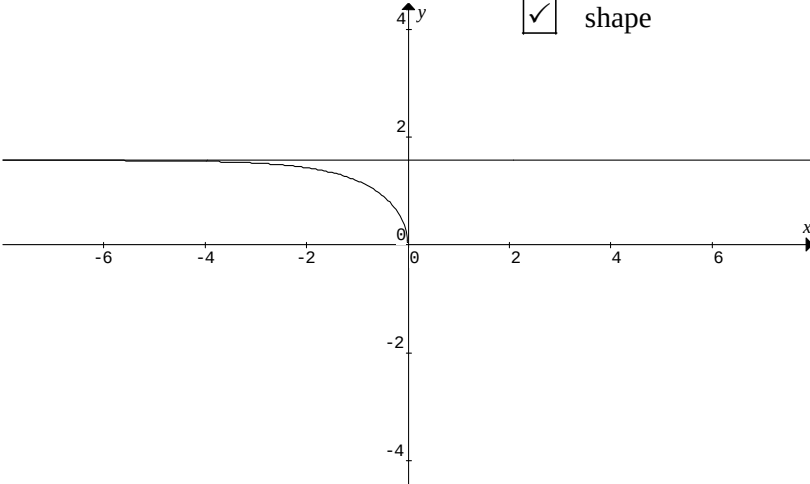
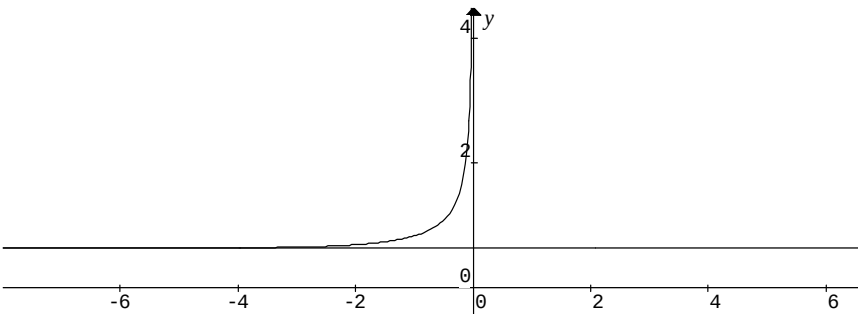
| Question | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks | Bands |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| 5(a)(i)  | $\angle BAC = 60$ ( $\triangle ABC$ is equilateral $\triangle$ )<br>$\angle BXC = 120$ (opposite angles in cyclic quad ABXC are supplementary)<br>$\angle AXC = \angle ABC = 60$ ( $\angle$ 's on same arc at circumference are equal)<br>$\therefore \angle AXC = \angle AXB = 60$<br><br>in $\triangle BDX$ and $\triangle ACX$<br>$\angle DXC = \angle AXB = 60$ (proved above) <input checked="" type="checkbox"/><br>$\angle DXB = \angle CAX$ ( $\angle$ 's at circumference on same arc) <input checked="" type="checkbox"/><br>$\therefore \triangle BDX \parallel \triangle ACX$ (equiangular) <input checked="" type="checkbox"/>                                                              | 3     |       |
| 5(a)(ii) | $\triangle CDX \parallel \triangle ABX$ (as proved in (i) above)<br>since $\triangle BDX \parallel \triangle ACX$<br>$\therefore \frac{BD}{AC} = \frac{BX}{AX} = \frac{DX}{CX}$ and $\frac{CD}{AB} = \frac{CX}{AX} = \frac{DX}{BX}$<br>$\therefore BD = \frac{BX \cdot AC}{AX}$ and $CD = \frac{CX \cdot AB}{AX}$ <input checked="" type="checkbox"/><br><br>since $BC = BD + DC$<br>hence $BC = \frac{BX \cdot AC}{AX} + \frac{CX \cdot AB}{AX}$ <input checked="" type="checkbox"/><br><br>$BC \cdot AX = (BX \cdot AC) + (CX \cdot AB)$<br>and as $BC = AC = AB$ (equilateral $\triangle$ )<br>$\therefore \div$ LHS and RHS by $BC$ <input checked="" type="checkbox"/><br>$\therefore AX = BX + CX$ | 3     |       |
| 5(b)(i)  | $y = \sin 9x$ when $0 < x < \pi$ it has $4\frac{1}{2}$ cycles,<br>more area above $x$ -axis than below<br>$\therefore \int_0^{\pi} \sin 9x \, dx \neq 0$ (false) <input checked="" type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |       |
| 5(b)(ii) | $x \sin x$ is an even function <input checked="" type="checkbox"/><br>$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin x \, dx = 2 \int_0^{\frac{\pi}{2}} x \sin x \, dx \neq 0$ <input checked="" type="checkbox"/><br>(false)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2     |       |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 5(c) | <p>slope tangent = <math>\frac{dy}{dx}</math> : ( differentiating implicitly)</p> $-2 \sin 2x + \cos y \frac{dy}{dx} = 0$ $\cos y \frac{dy}{dx} = 2 \sin 2x$ $\frac{dy}{dx} = \frac{2 \sin 2x}{\cos y} \quad \boxed{\checkmark}$ <p>At <math>(\frac{\pi}{6}, \frac{\pi}{6})</math>, slope of tangent = 2 <span style="float: right;"><math>\boxed{\checkmark}</math></span></p> <p>Equation of tangent is</p> $y - \frac{\pi}{6} = 2(x - \frac{\pi}{6})$ $y = 2x - \frac{\pi}{6} \quad \boxed{\checkmark}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3 |  |
| 5(d) | <p>Let <math>x = a \sin \theta \quad dx = a \cos \theta d\theta</math></p> $\int \sqrt{a^2 - x^2} dx = \int \sqrt{a^2 - a^2 \sin^2 \theta} a \cos \theta d\theta = a^2 \int \cos^2 \theta d\theta \quad \boxed{\checkmark}$ $= \frac{a^2}{2} \int \cos 2\theta + 1 d\theta =$ $\frac{a^2}{2} \left[ \frac{1}{2} \sin 2\theta + \theta \right] + C = \frac{a^2}{2} [\sin \theta \cos \theta + \theta] + C \quad \boxed{\checkmark}$ <p>From this triangle :</p> <div style="display: flex; align-items: center; justify-content: center;">  <div style="margin-left: 20px;"> <math>\frac{x}{a} = \sin \theta</math><br/> <math>\cos \theta = \frac{\sqrt{a^2 - x^2}}{a}</math><br/> <math>\theta = \sin^{-1} \frac{x}{a}</math> </div> </div> $\frac{a^2}{2} [\sin \theta \cos \theta + \theta] + C =$ $\frac{a^2}{2} \left[ \frac{x}{a} \frac{\sqrt{a^2 - x^2}}{a} + \sin^{-1} \frac{x}{a} \right] + C = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C \quad \boxed{\checkmark}$ | 3 |  |

| Question  | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks                                                                                                    | Bands |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------|
| 6(a)(i)   | $\overline{a\alpha^2 + b\alpha + c} = \overline{0}$<br>$\overline{a\alpha^2 + b\alpha + c} = 0$<br>$a\overline{\alpha^2} + b\overline{\alpha} + \overline{c} = 0$<br>$a\overline{\alpha}^2 + b\overline{\alpha} + \overline{c} = 0$<br>$\therefore \overline{\alpha}$ is a solution                                                                                                                                                                      | <b>1</b><br><br><br><br><br><br><br><br><br><br><b>1</b>                                                 |       |
| 6(a)(ii)  | $(x - 3i)(x + 3i)(x - (1 + 2i))(x - (1 - 2i))$<br>$(x^2 + 9)(x^2 - 2x + 5)$<br>$x^4 - 2x^3 + 14x^2 - 18x + 45$                                                                                                                                                                                                                                                                                                                                           | <b>1</b><br><br><br><br><br><br><br><b>1</b><br><b>1</b>                                                 |       |
| 6(b)      | $\cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$<br>$= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$<br>equate real part:<br>$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$<br>$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$<br>$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1$              | <b>1</b><br><br><br><br><br><br><br><b>1</b><br><br><br><br><br><b>1</b>                                 |       |
| 6(c)(i)   | $z^n = \cos n\theta + i \sin n\theta$<br>$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$<br>$= \cos n\theta - i \sin n\theta$<br>$z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$<br>$= 2 \cos n\theta$                                                                                                                                                                                                                       | <b>1</b><br><br><br><br><br><br><br><br><br><br><b>1</b>                                                 |       |
| 6(c)(ii)  | $\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$<br>$= \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$<br>$= 2 \cos 4\theta + 4(2 \cos 2\theta) + 6$<br>$\left(z + \frac{1}{z}\right)^4 = (2 \cos \theta)^4 = 16 \cos^4 \theta$<br>$\therefore 16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$<br>$\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$ | <b>1</b><br><br><br><br><br><br><br><br><br><br><b>1</b><br><br><br><br><br><b>1</b>                     |       |
| 6(c)(iii) | $\int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} d\theta$<br>$= \left[ \frac{1}{8} \frac{\sin 4\theta}{4} + \frac{1}{2} \frac{\sin 2\theta}{2} + \frac{3}{8} \theta \right]_0^{\frac{\pi}{2}}$<br>$= \frac{1}{32} \sin 2\pi + \frac{1}{4} \sin \pi + \frac{3\pi}{16} - 0$<br>$= \frac{3\pi}{16}$                                                                 | <b>1</b><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><b>1</b> |       |

| Question | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks                               | Bands |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------|
| 7(a)     | <p>Let roots be: <math>\alpha - d, \alpha, \alpha + d</math></p> <p>sum of roots: <math>3\alpha = -a \Rightarrow \alpha = -\frac{a}{3}</math></p> <p><math>\alpha = -\frac{a}{3}</math> is a root to: <math>x^3 + ax^2 + bx + c = 0</math></p> $\left(-\frac{a}{3}\right)^3 + a\left(-\frac{a}{3}\right)^2 + b\left(-\frac{a}{3}\right) + c = 0$ $-\frac{a^3}{27} + \frac{a^3}{9} - \frac{ab}{3} + c = 0$ $-a^3 + 3a^3 - 9ab + 27c = 0$ $2a^3 - 9ab + 27c = 0$                                                                                                                                                                                    | <p>1</p> <p>1</p> <p>1</p> <p>1</p> |       |
| 7(b)(i)  | <p>Consider the intersection of the two curves:</p> $y = \frac{c}{x^2}$ $\frac{x^2}{a^2} + \frac{c^4}{x^2 b^2} = 1 \quad x^4 b^2 - x^2 a^2 b^2 + a^2 c^4 = 0$ <p>Solving for <math>x^2</math>: <math>\Delta = a^4 b^4 - 4a^2 b^2 c^4</math></p> <p>for the roots to be real and distinct:</p> $\Delta > 0$ $a^4 b^4 - 4a^2 b^2 c^4 > 0$ $a^2 b^2 > 4c^2 \quad \text{or} \quad ab > 2c^2$ <p>If <math>ab &gt; 2c^2</math>, <math>x^2</math> has two distinct values and hence <math>x</math> has 4 values corresponding to 4 points of intersection.</p>                                                                                           | <p>1</p> <p>1</p> <p>1</p>          |       |
| 7(b)(ii) | $y = \frac{c^2}{x} \quad xa \sin \theta - \frac{c^2}{x} b \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$ $x^2 a \sin \theta - c^2 b \cos \theta = (a^2 - b^2) x \sin \theta \cos \theta$ $x^2 a \sin \theta - (a^2 - b^2) x \sin \theta \cos \theta - c^2 b \cos \theta = 0$ $\Delta = \left[ (a^2 - b^2)^2 \sin \theta \cos \theta \right]^2 + 4ac^2 b \cos \theta \sin \theta$ <p>If <math>0 &lt; \theta &lt; \frac{\pi}{2}</math>, <math>0 &lt; \sin \theta &lt; 1</math>, <math>0 &lt; \cos \theta &lt; 1</math>, <math>a, b &gt; 0</math></p> <p><math>\therefore \Delta &gt; 0</math> and this gives two values for <math>x</math>.</p> | <p>1</p> <p>1</p> <p>1</p>          |       |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |  |
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| 7(b)(iii) | $x_1 + x_2 = \frac{2(a^2 - b^2)\sin\theta\cos\theta}{2a\sin\theta} = \frac{(a^2 - b^2)\sin\theta\cos\theta}{a\sin\theta}$ $\frac{x_1 + x_2}{2} = \frac{(a^2 - b^2)\sin\theta\cos\theta}{2a\sin\theta}$ $x = \frac{(a^2 - b^2)\cos\theta}{2a} \quad (1)$ <p>sub into normal to find y:</p> $a \frac{(a^2 - b^2)\cos\theta}{2a} \sin\theta - yb\cos\theta = (a^2 - b^2)\sin\theta\cos\theta$ $\frac{(a^2 - b^2)}{2} \sin\theta - yb = (a^2 - b^2)\sin\theta \quad \cos\theta \neq 0$ $y = \frac{-(a^2 - b^2)}{2b} \sin\theta \quad (2)$ | 1           |  |
| 7(b)(iv)  | <p>Eliminate <math>\theta</math>:</p> <p>From (1) <math>\cos\theta = \frac{2ax}{a^2 - b^2}</math></p> <p>From (2) <math>\sin\theta = -\frac{2by}{a^2 - b^2}</math></p> <p><math>\sin^2\theta + \cos^2\theta = 1</math></p> $\frac{4a^2x^2}{(a^2 - b^2)^2} + \frac{4b^2y^2}{(a^2 - b^2)^2} = 1$ $\frac{x^2}{\left(\frac{a^2 - b^2}{2a}\right)^2} + \frac{y^2}{\left(\frac{a^2 - b^2}{2b}\right)^2} = 1$                                                                                                                                | 1<br>1<br>1 |  |

| Question  | Criteria                                                                                                                                                                                                                                                                                                                                                       | Marks | Bands |
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| 8(a)(i)   | <p>Domain: <math>e^x &gt; 0</math> for all <math>x</math></p> <p>for <math>\cos^{-1}(e^x): -1 \leq e^x \leq 1</math> only if <math>0 \leq e^x \leq 1</math>, i.e. if <math>x \leq 0</math>. <input checked="" type="checkbox"/></p> <p>Range: For this domain range will be : <math>0 \leq y \leq \frac{\pi}{2}</math> <input checked="" type="checkbox"/></p> | 2     |       |
| 8(a)(ii)  | <p><input checked="" type="checkbox"/> limit at <math>y = 1</math></p> <p><input checked="" type="checkbox"/> shape</p>                                                                                                                                                     | 2     |       |
| 8(a)(iii) |  <p><input checked="" type="checkbox"/> limit at <math>y = 1</math></p> <p><input checked="" type="checkbox"/> shape (reciprocal of (ii))</p>                                                                                                                              | 2     |       |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 8(b)(i)  | <p>prove true for <math>n = 1</math></p> <p><math>\therefore (1 + \sqrt{2})^1 = 1 + \sqrt{2}</math> true where <math>p_n = 1</math> and <math>q_n = 1</math> <input checked="" type="checkbox"/></p> <p>assume true for <math>n = k</math></p> <p><math>\therefore (1 + \sqrt{2})^k = p_k + q_k \sqrt{2}</math></p> <p>prove true for <math>n = k + 1</math></p> <p><math>\therefore (1 + \sqrt{2})^{k+1} = (1 + \sqrt{2})^k (1 + \sqrt{2})^1</math></p> <p><math>= (p_k + q_k \sqrt{2})(1 + \sqrt{2})</math> (by assumption above)</p> <p><math>= p_k + p_k \sqrt{2} + q_k \sqrt{2} + 2q_k</math></p> <p><math>= (p_k + 2q_k) + (p_k + q_k) \sqrt{2}</math> <input checked="" type="checkbox"/></p> <p>since <math>p_k</math> and <math>q_k</math> are integers</p> <p><math>\therefore p_k + 2q_k</math> is an integer <math>= p_{k+1}</math></p> <p><math>\therefore p_k + q_k</math> is an integer <math>= q_{k+1}</math></p> <p>hence <math>(1 + \sqrt{2})^{k+1} = p_{k+1} + q_{k+1} \sqrt{2}</math> <input checked="" type="checkbox"/></p> <p>If true for <math>n = k</math> and <math>n = k + 1</math> and since true for <math>n = 1, 2, 3, \dots</math></p> <p><math>\therefore</math> true for <math>\forall n</math> positive integers</p> | 3 |  |
| 8(b)(ii) | <p><math>p_1^2 - 2q_1^2 = 1 - 2 \times 1^2 = -1 = (-1)^1</math></p> <p>If <math>p_k^2 - 2q_k^2 = (-1)^k</math></p> <p>then when <math>n = k + 1</math></p> <p><math>(p_{k+1})^2 - (q_{k+1})^2 = (p_k + 2q_k)^2 - 2(p_k + q_k)^2</math> (from above) <input checked="" type="checkbox"/></p> <p><math>= p_k^2 + 4p_k q_k + 4q_k^2 - 2p_k^2 - 4p_k q_k - 2q_k^2</math></p> <p><math>= 2q_k^2 - p_k^2</math></p> <p><math>= -1(p_k^2 - 2q_k^2)</math></p> <p><math>= -1 \times (-1)^k</math></p> <p><math>= (-1)^{k+1}</math> <input checked="" type="checkbox"/></p> <p>if true for <math>n = k</math> and <math>n = k + 1</math></p> <p>and since true for <math>n = 1, 2, 3, \dots</math></p> <p><math>\therefore</math> true for <math>\forall n</math> positive integers</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2 |  |
| 8(c)(i)  | <p><math>f(xa) = f(x) + f(a)</math></p> <p>if <math>x = 1</math> then <math>f(a) = f(1) + f(a) \Rightarrow f(1) = 0</math></p> <p>if <math>x = a</math> then <math>f(a^2) = f(a) + f(a)</math></p> <p><math>f(a^2) = 2f(a)</math> <input checked="" type="checkbox"/></p> <p><math>\therefore 2f(-1) = f(-1^2) = f(1) = 0</math> <input checked="" type="checkbox"/></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2 |  |
| 8(c)(ii) | <p>since <math>2f(a) = f(a^2)</math></p> <p><math>\therefore 2f(-a) = f((-a)^2)</math></p> <p><math>= f(a^2)</math></p> <p><math>= 2f(a)</math> <input checked="" type="checkbox"/></p> <p>since <math>2f(a) = 2f(-a)</math></p> <p><math>\therefore f(a) = f(-a)</math> even function <input checked="" type="checkbox"/></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2 |  |